NEURALPATH CHEATSHEET

Module 9: AI Safety, Ethics & Interpretability

Audit fairness, compute SHAP/LIME attributions, inspect Sparse Autoencoders (SAEs), defend agai

Core Formulas, Mental Models & Engineering Rules

- Fairness trade-offs: Demographic Parity vs. Equalized Odds vs. Calibration.
- SHAP computes unique game-theoretic additive feature attributions summing to f(x) - E[f(x)].
- Mechanistic Interpretability: Sparse Autoencoders (SAEs) isolate monosemantic features.
- Defend agents against indirect prompt injection with least-privilege tool allowlists & HITL.

```
Estimated study time: 4 hours · Badge: Responsible AI Steward
```